LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING, (AUTONOMOUS)
L.B.Reddy Nagar :: Mylavaram – 521 230 :: Krishna Dist.

M.Tech I Semester ~~Regular/~~Supplementary Examinations

MEE102-POWER CONVERTERS 1 APR 2015

Time : 3 hours Max. Marks : 60

Answer all the questions
All Questions carry equal Marks
****

1(a) For a single phase a.c voltage regulator feeding a resistive load, draw the waveforms of source voltage, gating signals, output voltage, source and output currents and voltage across SCRs. Describe its working with reference to the waveforms drawn. [5M]

(b) For a single phase a.c regulator feeding a resistive load, show that power factor is given by the expression $\left[\frac{1}{\pi}[\pi-\alpha]+\frac{\sin 2\alpha}{2}\right]^{\frac{1}{2}}$ [7M]

OR

(c) A 3-star connected balanced resistances are supplied from a 3-phase AC voltage controller. Derive the expression for R.M.S value of load current in the complete range of firing angles. Draw the waveforms of load current. [12M]

2(a) Evaluate the input power factor and harmonic factors for a three phase half Controlled converters. [6M]

(b) Explain the operation of a three phase twelve pulse converter along with necessary circuit diagram and waveforms [6M]

OR

(c) Explain in detail the extinction angle control and symmetric angle control methods used for the improvement of power factor of phase controlled converters. [12M]

3(a) What are the various power factor improvement methods in converters? Discuss and analyze the need for power factor correction/improvement in converters. [12M]

OR

(b) How can the input current of the rectifier-fed boost converter be made sinusoidal and in phase with the input voltage? Explain in detail. [12M]

4(a) Explain briefly the following modulation techniques with relative advantages and disadvantages.
i) Multiple PWM ii) Sinusoidal PWM iii) Delta modulation. [12M]

OR

(b) Explain about the phase displacement control and advanced harmonic injection modulation of a single phase inverter. [12M]

5(a) What are Multi-Level Inverters? Explain with circuit diagram switching of Multi-level inverters. State its advantages & disadvantages. [12M]

OR

(b) List out the advantages and disadvantages of flying capacitor multilevel inverter. [6M]

(c) Discuss the basic concept and features of multilevel inverter [6M]

LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING, (AUTONOMOUS)
L.B.Reddy Nagar :: Mylavaram – 521 230 :: Krishna Dist.

M.Tech I Semester ~~Regular~~/Supplementary Examinations

MEC102-TRANSFORM TECHNIQUES 11 APR 2015

Time : 3 hours Max.Marks :60

Answer all the questions
All Questions carry equal Marks
****

1(a) Define Fourier transform pair and define conditions for its existence and hence explain the frequency analysis of signals with the help of Fourier transform. [6M]
(b) Discuss about Walsh and Haar transforms. [6M]

(OR)

(c) Explain about Hilbert transform properties and applications? [6M]
(d) Explain the relation between Fourier and z-transform techniques. [6M]

2(a) What are the drawbacks of FT? How these can be overcome using STFT? What are the properties of STFT? [6M]
(b) What is wavelet? Discuss clearly the features of wavelet transform and hence distinguish between continuous and discrete wavelet transforms. [6M]

(OR)

(c) Define MRA scaling function and explain the concept of basis for wavelet subspace using Haar wavelet . [6M]
(d) Discuss about
(i) Mexican Hat Meyer (ii) Doubechies wavelet transforms [6M]

3(a) Define orthogonal wavelets and bring out their relation to filter banks. [6M]
(b) Discuss the implementation of decomposition filters and hence bring out concept of PRQMF. [6M]

(OR)

(c) What are the drawbacks of FT? How these can be overcome using STFT? What are the properties of STFT? [6M]
(d) Show that CWT and ICWT form wavelet transform pair and Explain DWT with mathematical equations. [6M]

4(a) Explain about multi wavelets? [6M]
(b) What do you understand by Bi-orthogonal filters? [6M]

(OR)

(c) Write short notes on B-Splines? [6M]
(d) Discuss about wavelet packet transform? [6M]

5(a) What are wavelet packets? [6M]
(b) Explain about signal compression using DWT and 2-D DWT? [6M]

(OR)

(c) Explain about sub band coding of speech and music? [6M]
(d) What are the various signal compression techniques and explain? [6M]

****

Code: MEC104

LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING (AUTONOMOUS)
L.B. Reddy Nagar :: Mylavaram – 521 230 :: Krishna Dist.

M.Tech. I Semester Regular/Supplementary Examinations

MICROCONTROLLERS FOR EMBEDDED SYSTEM DESIGN
(Systems and Signal Processing (ECE))

Time : 3 hours Max. Marks: 60

Answer all the questions
All Questions carry equal marks 6 APR 2015

******

1(a) Explain the classification of Embedded system? [6M]
(b) Discuss about the design technology and challenges of an embedded system with an example? [6M]

(OR)

(c) Define the term embedded systems? and write the application of embedded system with examples? [6M]
(d) Explain different embedded Hardware units in detail? [6M]

2(a) With a diagram explain the interfacing processor of 8051 PIC? [6M]
(b) What is the importance of input/output ports and explain them? [6M]

(OR)

(c) Explain the differences between microcontrollers and microprocessors based on the applications? [6M]
(d) What is the importance to memory arbitration schemes? [6M]

3(a) Write the important differences between RISC & CISC processors? [6M]
(b) Explain about ARM processor architecture? [6M]

(OR)

(c) What is a programmable system-on-chip method explain it? [6M]
(d) Write about (i) Timer blocks (ii) Digital blocks [6M]

4(a) What is an interrupt and exception explain them? [6M]
(b) Explain the importance of device drivers for internal programmable timing devices? [6M]

(OR)

(c) Explain about different interrupt handling schemes with examples? [8M]
(d) Write a short notes on context switching? [4M]

5(a) Explain various types of bus architecture serial communication protocols? [6M]
(b) What are the different popular protocols used for wireless communication? [6M]

(OR)

(c) Explain how Bluethooth devices can be used to setup personnel area network? [6M]
(d) Write a short notes on Ethernet protocol? [6M]

^^^^^^^^^^^^^^^^